LeadAnchor

Requirements check, and the services the app uses

Checked against: "LeadAnchor_Specs Doc.pdf" (7 pages) and "Revenue_Automation_OS_Capabilities v2.xlsx" (51 capabilities, C-01 to C-52).
Date: 28 October 2026. **Based on:** the code on this computer today. The app is not online, and nothing has been saved to git yet.

1. The short answer

4	4	12	27	4
Fully done	Built, only waiting for a Telnyx account	Partly done	Not started	Deferred by the spec itself

The core of the product works today. A missed call is caught. The AI Dispatcher talks to new callers and asks if it can text them. The customer gets one text with a form link, sends details and photos, and the owner sees the job on the lead board and the money on the Today page. This is Phase 1 ("Core capture loop") in the spec PDF and the "Pilot" phase in the Excel.

Everything about money, booking, Google, campaigns and memberships has not been started. That is Phases 2 to 4 in the spec PDF and MVP1/MVP2 in the Excel.

Only one outside account exists: Google AI Studio, for the Gemini AI, on the free plan. Every other service (Telnyx, WhatsApp, Firebase, Stripe, Google Business Profile, email) has no account yet. See section 6.

> **What "simulator" means.** Telnyx is the phone company the spec requires. Without a Telnyx account the app can't make or receive real calls or texts. So the app has a built-in **call simulator** that plays the caller and the phone network. The rest of the app runs the same code it will run with Telnyx. Only the small Telnyx connection part is swapped.

Progress by Excel phase

Excel phase	Capabilities	Fully done	Built, needs account	Partly done	Not started
Pilot (first 6–8 weeks)	14	4	3	6	1
Pilot + MVP1	15	0	1	6	8
MVP1 (months 3–6)	7	0	0	0	7
MVP2 (months 7–12)	11	0	0	0	11
Phase 2 (year 2+)	4	Deferred by the Excel itself ("DEFERRED — privacy")			

13 of the 14 Pilot capabilities have been started. The only Pilot item not started is C-10 (website and Google "Book / Chat" button).

2. How to read the statuses

Status	Meaning
Fully done	Works today. Nothing important is missing.
Built, needs account	All of our logic is built and works on the call simulator. It goes live once a Telnyx account is created and connected. It still needs one test with real calls.
Partly done	Some of it works. The missing pieces are listed next to it.
Not started	No work done yet. Most of these belong to later phases.
Deferred by spec	The Excel marks it “Phase 2 — deferred”, and the spec PDF leaves its “Future” phase out of the estimate. Not expected now.

3. Excel capabilities, grouped by status

Fully done 4 capabilities

ID	Capability	Phase	What works
C-04	Know who’s calling instantly	Pilot	New callers go to the AI Dispatcher. Returning customers hear the voicemail greeting, and the owner gets a “Missed call from <name>” alert. The lead card shows who the caller is.
C-08	Simple form customers fill in	Pilot	A mobile form with no login: pick a service, describe the job, add photos, name, address and a texting-permission box. It creates a clean, structured lead.
C-09	Customers send a photo	Pilot	Camera or gallery upload, with a thumbnail made automatically. The Gemini AI reads the item, brand and model (only from labels it can read), urgency and visible safety risks. Notes: photos are saved on this computer for now (online storage is needed when the app goes live). The AI part pauses when the free daily AI limit runs out.
C-12	Your simple lead board	Pilot	Columns New, Contacted, Quoted, Scheduled, Won and Lost. Drag cards on a computer; use the move button on a phone. Each card has Call and Text buttons. Won asks for the real job value, Lost asks for a reason.

Built, needs account 4 capabilities

ID	Capability	Phase	What is built	What is left
C-02	Your own smart business number	Pilot	Every business gets its own number at sign-up. The code that orders real numbers from Telnyx is written.	Today the simulator hands out made-up 555 numbers. Needs a Telnyx account, then a test that a new number works within 60 seconds.
C-03	Catch every incoming call	Pilot	The receiver for Telnyx call events is built. It checks Telnyx’s digital signature, ignores repeats and copes with events arriving out of order. It has automated tests.	Needs a Telnyx account and a public internet address so Telnyx can reach the app. Then a real-call test (the spec asks for the call to be logged within 2 seconds).

ID	Capability	Phase	What is built	What is left
C-13	Voicemails turned into leads	Pilot + MVP1	Returning callers hear a voicemail greeting. The message is recorded and turned into text, and the AI pulls out what the caller needs.	Recording and speech-to-text are Telnyx features, so this is proven only on the simulator.
C-39	AI Dispatcher	Pilot	Answers only new callers. Says the business name and that it is an automated assistant. Asks one question, then "Can we text you?", and saves the spoken "yes" with its time and words before any text goes out. The server ends the call at 60 seconds. Then it sends the form link.	Real voice needs Telnyx. The spec's "sub-second, streaming" voice is not built: the app uses Telnyx's simpler speak-and-listen commands. If a caller keeps talking, listening stops at 20 seconds and the AI moves on.

Partly done **12 capabilities**

ID	Capability	Phase	What works	What is missing
C-01	Set up missed-call catching	Pilot	Settings shows the call-forwarding codes for Verizon, AT&T, T-Mobile and other carriers. On Android, tapping a code opens the dialer with it filled in; on iPhone it can be copied. Or the owner can just use the LeadAnchor number.	The carrier isn't detected automatically (the owner picks it). No test call confirms forwarding works.
C-05	Never lose a hang-up	Pilot	A hang-up creates a lead marked "Waiting", and a "Missed call… tap to call back" alert appears in the app within about a second.	No real phone notification (Firebase / Apple push). The alert only shows while the app is open. The sending code exists; the Firebase project and the browser's notification sign-up are missing.
C-06	Auto-greeting that captures leads	Pilot	The greeting and the recorded permission work through the AI Dispatcher, with a spoken "yes".	The "press 1" keypad option isn't built. The newer C-39 asks for a spoken yes instead, so "press 1" may no longer be needed. Worth confirming with the manager.
C-07	Auto-text the booking link	Pilot + MVP1	After a "yes", exactly one text with a personal form link is sent through the single governed send path. Texts are blocked until carrier approval. Opening the link is recorded.	Real SMS needs Telnyx. The Day-1 WhatsApp or web route (for before carrier approval) is not wired up.
C-11	Chat on WhatsApp	Pilot	Code to send and receive WhatsApp messages, with Meta's signature check and STOP/START handling. A builder for the wa.me chat link.	No Meta WhatsApp account. The AI doesn't reply to chats. The WhatsApp link isn't shown anywhere yet.
C-14	See the money you recovered	Pilot	Today page: money recovered (real job values only), calls caught, open jobs, leads won, 7/30/90-day views, a daily chart and a plain-English summary.	The urgency breakdown isn't shown.

ID	Capability	Phase	What works	What is missing
C-17	Set your services & prices	Pilot + MVP1	Each business starts with a service list for its trade. Customers pick from it on the form, and the AI uses it. The database already has fields for duration and price.	No screen for the owner to add, edit or price services.
C-27	Built-in permission tracking	Pilot + MVP1	A permanent, tamper-evident record of every yes and no: phone, type, time, IP address (web form), source, the exact wording version and which business. It is checked before every text. STOP and START work instantly. Lists can't be uploaded.	The spec names a Firebase ledger. That code is written, but there is no Firebase project, so the record lives in our own database for now. The form has one permission box; the spec asks for two separate boxes (job texts and marketing).
C-34	Customer history view	Pilot + MVP1	Each lead page shows the customer, the full activity history, photos and job value. The search box finds customers.	No single customer page that combines all their jobs, payments and lifetime value.
C-40	Value-proof notification	Pilot + MVP1	The summary card on the Today page (real numbers only), and the daily / weekly / off setting.	Nothing sends the summary automatically yet. There is no scheduled job and no push.
C-41	Licensed & insured profile	Pilot + MVP1	Settings collects licence number, insurance and years in business. The customer form shows "Licensed & Insured", never "Verified".	No profile card in the first text, public profile page, Google rating, work-photo gallery or per-message on/off switch.
C-43	Carrier registration (10DLC)	Pilot	Two paths: business with an EIN, or sole proprietor. The EIN format is checked. Tax ID and birth date are encrypted and only shown masked. Status appears in plain English. All SMS is blocked until approval.	Nothing is sent to the carriers yet. The Telnyx registration steps (brand, campaign, number), status checks, resubmission and the sole-proprietor phone code (OTP) aren't built. Approval is faked with a developer button. No data-retention policy or privacy-policy wording yet.

Not started 27 capabilities

ID	Capability	Phase	Main outside service it will need
C-10	Capture from your website & Google	Pilot	Website widget, Google Business Profile
C-15	Take a deposit upfront	Pilot + MVP1	Stripe
C-16	An assistant that answers for you (text AI chat)	Pilot + MVP1	AI (Gemini), SMS
C-18	Let customers book themselves	Pilot + MVP1	Calendar or booking tool, Google
C-19	Automatic booking confirmations	Pilot + MVP1	SMS
C-20	Reminders that cut no-shows	Pilot + MVP1	SMS, booking tool
C-21	Keep Google up to date automatically	Pilot + MVP1	Google Business Profile
C-33	Website + better Google listing (add-on)	Pilot + MVP1	Google Business Profile, WhatsApp
C-45	Turn a photo into a quote in one tap	Pilot + MVP1	AI, PDF maker, Stripe

ID	Capability	Phase	Main outside service it will need
C-22	Fill cancelled slots automatically	MVP1	SMS, booking tool
C-23	Turn happy jobs into more reviews	MVP1	Google Business Profile
C-24	Pass extra jobs to trusted pros	MVP1	Other businesses on LeadAnchor, partner terms
C-25	Earn credits for referrals	MVP1	Stripe Billing
C-26	Get paid faster after the job	MVP1	Stripe
C-46	Compliant review request (two links)	MVP1	Email, Google
C-47	AI replies to every Google review	MVP1	Google Business Profile, AI
C-28	Win back old customers	MVP2	Email now, SMS later
C-29	Run a promotion in minutes	MVP2	Email now, SMS later
C-30	Fill slow days with weather and holiday offers	MVP2	Weather and holiday data
C-31	Let customers opt into offers (keyword)	MVP2	WhatsApp
C-42	Get found when customers ask AI for a plumber	MVP2	Public pages, AI services
C-44	Customers reply to an offer and book	MVP2	Campaigns, booking
C-48	Build your own monthly maintenance plans	MVP2	Stripe Billing
C-49	Charge members automatically every month	MVP2	Stripe Billing
C-50	See your recurring income at a glance	MVP2	Stripe Billing
C-51	Paid work shows up on the board	MVP2	Stripe Billing
C-52	Members update or cancel without calling	MVP2	Stripe customer portal

Deferred by spec 4 capabilities

ID	Capability	Why it isn't expected now
C-35	Win recurring work from property managers	The Excel marks all four "Phase 2 — DEFERRED (privacy)": they share data between businesses. The spec PDF puts them in Phase 5 ("Future") and says leaving them out of the estimate is fine.
C-36	Get jobs from warranty companies	
C-37	A marketplace that sends you customers	
C-38	A stronger network in your area	

4. Spec PDF check

4.1 The 14 functional modules

#	Module	Status	Notes
1	Telephony & call handling	Built, needs account	The call flow is built: ring the owner, then the AI or voicemail, plus hang-up handling. One number per business; safe handling of call events. Real calls need Telnyx.

#	Module	Status	Notes
2	Conversational voice layer	Built, needs account	A short scripted AI conversation with a 60-second limit. Real voice needs Telnyx. The low-latency streaming voice isn't built.
3	Intake & media capture	Fully done	Mobile form, photo upload and AI photo reading.
4	Work management	Partly done	Board, stages and per-lead history are done. Missing: the second board view, a per-customer history screen and tags.
5	Messaging orchestration	Partly done	The single governed send path, with its rules, is done. SMS is built (simulated). WhatsApp code is built, with no account. RCS and email aren't built. Incoming STOP/START is done.
6	Documents & quoting	Not started	
7	Payments	Not started	
8	Scheduling	Not started	
9	Listings integration (Google)	Not started	
10	Reporting	Partly done	The headline dashboard and the call list are done. The fixed set of reports isn't built.
11	Outbound campaigns	Not started	Marketing texts are blocked on purpose by the send rules (the Excel's email-first plan).
12	Recurring billing	Not started	
13	Compliance infrastructure	Partly done	Permission capture and enforcement are done, and the consent and audit records are tamper-evident. Automatic carrier registration isn't built (only the data collection and the SMS block). Only a few owner actions are written to the audit log so far.
14	Operator console	Not started	No internal tool yet for onboarding help, support, managing businesses or monitoring.
	Public web presence	Not started	No marketing site and no public page per business.

4.2 Delivery phases

Phase	Status	Notes
0 Foundations	Mostly done	Done: architecture, keeping each business's data separate (enforced inside the database), sign-in, design system, governed send path, audit logging. Missing: CI/CD (automatic build, test and deploy) and separate environments (only this computer).
1 Core capture loop	Mostly built	Modules 1, 2, 3, part of 4 and part of 13 work on the simulator. Not yet "production-grade": the app isn't online, Telnyx isn't connected, and carrier registration isn't automated.
2 Workflow & money	Not started	Only the board from module 4 exists.
3 Growth features	Not started	Only the headline dashboard exists.
4 Monetisation & scale	Not started	

Phase	Status	Notes
5 Future	Out of scope	The spec says leaving it out is fine.

> **Most important gap.** The spec calls carrier registration the “critical sequencing constraint”. It takes 1 to 4 weeks per business, can’t be sped up, and must be automated through the Telnyx API from the start. Today we collect the data and block texts until approval, but we don’t submit anything to the carriers. This should be the first thing built once a Telnyx account exists.

4.3 Quality requirements

Requirement	Status	Notes
Mobile-first	Fully done	Designed for phones first: bottom tabs, bottom sheets and large tap targets. Also works on a computer.
Availability	Not started	Runs only on this computer. Needs reliable hosting, especially for call capture.
Speed	Partly done	Pages feel fast here, but they haven’t been measured on mobile networks with the app online. The voice path isn’t the streaming design the spec asks for.
Tamper-evident records	Fully done	Consent and audit records are chained together, so any edit or deletion is detectable. The database account the app uses can’t change or delete them.
Data protection	Partly done	Tax ID and birth date are encrypted and masked. No card data is stored. Missing: a data-retention policy. Whole-database encryption will come from the hosting provider.
English and Spanish	Not started	English only. The AI Dispatcher does understand “sí” as yes.
Monitoring	Partly done	Detailed logs only. No tracing and no alerts.
Swappable providers	Fully done	Phone, AI, messaging, push, photo storage and the consent ledger each sit behind a swappable connector. For example, Gemini could be replaced without touching business logic.

4.4 Ways of working and “not in scope”

- **Own the source repository from the first commit:** Partly done A git repository exists (branch “feat/foundation”), but nothing has been committed yet and it isn’t on GitHub or a similar site.
- **Not in scope:** Respected No native mobile app, page builder, accounting link, BI tool or non-US market was built, as the spec asks.

5. What the application is built with

Everything below is free and open source. Nothing here needs an account.

Part	Tool	What it does
Web app what owners and customers see	Next.js 16 and React 19	The web app itself: pages, navigation, loading data.
	Tailwind CSS 4, shadcn/ui (Base UI)	The look and the ready-made, accessible building blocks (buttons, sheets, menus).
	Tabler Icons; Archivo and Atkinson Hyperlegible Next fonts	Icons and easy-to-read fonts.
	TanStack Query	Fetches data from the server and keeps it up to date.
	dnd-kit (Dice UI Kanban)	Drag and drop on the lead board.
	Recharts, NumberFlow	The dashboard chart and the animated numbers.
	React Hook Form, Zod, Motion, Sonner, cmdk, nuqs, next-themes, date-fns	Forms and input checks, animations, pop-up alerts, the search box, remembering choices in the page address, dark/light mode, and dates.
Server the "brain" behind the app	Node.js 22 and TypeScript 5.9	The language and engine for all the code.
	Fastify 5	The API server the web app talks to.
	Better Auth	Sign-up and sign-in. Runs inside our app; passwords stay in our database.
	PostgreSQL 18	The database. On this computer it runs on its own, without Docker.
	Drizzle ORM	Reads and writes the database safely.
	pg-boss	Background jobs with automatic retries (AI work, sending messages).
	sharp	Resizes customer photos and makes thumbnails.
	Google Gen AI SDK, firebase-admin	Connectors for Gemini (in use) and Firebase (ready, not in use).
	pino, Vitest, libphonenumber-js	Logs, automated tests (34 today), and US phone-number handling.
Project tools	pnpm workspaces, Turborepo, Prettier	Keep the web app, server and shared code in one project, run them together, and keep the code tidy.

Safety built into the app

- Each business's data is walled off inside the database itself, not only by the app's code.
- Messages from Telnyx and WhatsApp are checked with their digital signatures before the app trusts them.
- API keys live only in a private settings file (.env) on the server. They never reach the browser.
- Tax IDs and birth dates are encrypted, and only the last digits are ever shown.
- Every outgoing message passes one rule check (permission, carrier approval, marketing block) before it is sent.

6. Outside services and accounts

Accounts created so far: 1 of 14. Only Google AI Studio (for Gemini) has an account and key. No account exists for Telnyx or any other service. Docker is not needed.

Service	What it's for	Required by	Code written?	Account?	Status today
Telnyx	Phone numbers, calls, call control, SMS, speech-to-text, text-to-speech	Spec (the Excel names Telnyx)	Yes: calls, numbers, SMS, voice. Not the carrier-registration steps.	**No**	The built-in call simulator is used instead.
The Campaign Registry (10DLC)	US carriers' permission to send texts	Spec	Partly: the form and stored data, not the submission	**No**	Done through the Telnyx account. Approval is faked with a developer button.
Google Gemini (Google AI Studio)	AI: what callers need, reading photos	Spec asks for AI; no vendor named. You chose Gemini.	Yes	**Yes**	**Live**, model gemini-3.8-flash, free plan: 20 requests a day per model. Today's limit is used up. Turning on billing removes the limit.
Meta WhatsApp Business Platform	WhatsApp chat with customers	Spec	Partly: send, receive, signature check	**No**	Not connected.
Firebase (Firestore and Cloud Messaging)	The consent ledger, and notifications to the owner's phone	Excel ("Firebase ledger", FCM)	Partly: ledger and notification sender	**No**	Consent is kept in our own database. Alerts only show inside the app.
Stripe (Connect and Billing)	Deposits, pay links, memberships; each business as a sub-account	Spec	No	**No**	Not started.
Google Business Profile	Google listing sync, reviews, "Book" button	Spec	No	**No**	Not started.
Google Calendar and Microsoft Outlook	Two-way calendar sync	Spec	No	**No**	Not started.
Booking tools (Calendly, Jobber, Acuity…)	Showing free slots and self-booking	Excel	No	**No**	Not started.
Email provider (for example Postmark, Resend, Amazon SES)	Transactional and marketing email	Spec	No	**No**	Not started. The app's Integrations page says emails are "kept in the local outbox", but no email is created or sent anywhere yet.

Service	What it's for	Required by	Code written?	Account?	Status today
RCS (branded rich texts)	Branded messages with plain-text fallback	Spec	No	**No**	Not started.
Weather and holiday data	Triggers for seasonal offers	Excel (MVP2)	No	**No**	Not started.
Online photo storage (for example Amazon S3, Cloudflare R2)	Keeping customer photos online	Excel ("cloud storage")	The connector is ready (a local folder today)	**No**	Photos are saved on this computer.
Hosting and online database	Putting the app on the internet, always on	Needed to go live (spec: high availability)	Not applicable	**No**	Runs only on this computer.

Suggested order to go live

1. **Create a Telnyx account** and buy a test number. Put its API key, public key and connection ID in the server's private settings file, and switch the phone and SMS settings from "simulator" to "telnyx".
2. **Put the app on the internet** (hosting, or a secure tunnel for testing) so Telnyx can reach it, then test real calls.
3. **Build the automatic carrier registration** through the Telnyx API. This is the spec's critical item, and each business waits 1 to 4 weeks for approval.
4. **Turn on billing for Gemini** so the AI doesn't stop after 20 requests a day.
5. **Create a Firebase project** for phone notifications and the Firebase consent ledger.
6. **Create a Meta WhatsApp Business account** for the Day-1 WhatsApp route.
7. **Save the code to git** and push it to the company's GitHub, then add CI/CD.
8. **Then the Phase 2 work:** Stripe, Google Business Profile, calendars and email.